

UNIVERSITY OF MADRAS

M.C.A. (DATA SCIENCE)

REGULATIONS ON LEARNING OUTCOMES-BASED CURRICULUM FRAMEWORK FOR POSTGRADUATE EDUCATION	
Programme	MCA
Programme Code	
Duration	PG - Two Years
Programme Outcomes (POs)	PO1: Problem Solving Skill Apply knowledge of Management theories and Human Resource practices to solve business problems through research in Global context. PO2: Decision Making Skill Foster analytical and critical thinking abilities for data-based decision-making. PO3: Ethical Value Ability to incorporate quality, ethical and legal value-based perspectives to all organizational activities. PO4: Communication Skill Ability to develop communication, managerial and interpersonal skills. PO5: Individual and Team Leadership Skill Capability to lead themselves and the team to achieve organizational goals. PO6: Employability Skill Inculcate contemporary business practices to enhance employability skills in the competitive environment. PO7: Entrepreneurial Skill Equip with skills and competencies to become an entrepreneur. PO8: Contribution to Society Succeed in career endeavors and contribute significantly to society. PO 9 Multicultural competence Possess knowledge of the values and beliefs of multiple cultures and a global perspective. PO 10: Moral and ethical awareness/reasoning Ability to embrace moral/ethical values in conducting one's life.
Programme Specific Outcomes (PSOs)	PSO1 – Placement To prepare the students who will demonstrate respectful engagement with others' ideas, behaviors, beliefs and apply diverse frames of reference to decisions and actions.

	SO 2 - Entrepreneur To create effective entrepreneurs by enhancing their critical thinking, problem solving, decision making and leadership skill that will facilitate startups and high potential organizations. PSO3 – Research and Development Design and implement HR systems and practices grounded in research that comply with employment laws, leading the organization towards growth and development. PSO4 – Contribution to Business World To produce employable, ethical and innovative professionals to sustain in the dynamic business world. PSO 5 – Contribution to the Society To contribute to the development of the society by collaborating with stakeholders for mutual benefit.
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**PROGRAMME OUTCOMES (PO) - PROGRAMME SPECIFIC OUTCOMES (PSO)
MAPPING**

PROGRAMME SPECIFIC OUTCOMES (PSO)					
	PO1	PO2	PO3	PO4	PO5
PSO1	3	3	3	3	3
PSO2	3	3	3	3	3
PSO3	3	3	3	3	3
PSO4	3	3	3	3	3
PSO5	3	3	3	3	3

Level of Correlation between PO's and PSO's

(Suggested by UGC as per Six Sigma Tool – Cause and Effect Matrix)

Assign the value

1 – Low

2 – Medium

3 – High

0 – No Correlation

METHODS OF EVALUATION		
Internal Evaluation	Continuous Internal Assessment Test	25 Marks
	Assignments / Snap Test / Quiz	
	Seminars	
	Attendance and Class Participation	
External Evaluation	End Semester Examination	75 Marks
Total		100 Marks
METHODS OF ASSESSMENT		
Remembering (K1)	- The lowest level of questions require students to recall information from the course content. - Knowledge questions usually require students to identify information in the text book.	
Understanding (K2)	- Understanding off acts and ideas by comprehending organizing, comparing, translating, interpolating and interpreting in their own words. - The questions go beyond simple recall and require students to combine at a together.	
Application (K3)	- Students have to solve problems by using/applying a concept learned in the class room. - Students must use their knowledge to determine a exact response.	
Analyze (K4)	- Analyzing the question is one that asks the students to break down something in to its component parts. - Analyzing requires students to identify reasons causes or motives and reach conclusions or generalizations.	
Evaluate (K5)	- Evaluation requires an individual to make judgment on something. - Questions to be asked to judge the value of an idea, a character, a work of art, or a solution to a problem. - Students are engaged in decision-making and problem –solving. - Evaluation questions do not have single right answers.	
Create (K6)	- The questions of this category challenge students to get engaged in creative and original thinking. - Developing original ideas and problem solving skills.	

Course	Number of Credits	Hours Per Week	Examination Duration (hrs)	Marks		
				I. A	ESE	Total
Semester - I						
Core - I Theory Mathematics for Data Science	4	5	3	25	75	100
Core - II Theory Advanced Data Structures and Algorithms	4	5	3	25	75	100
Core - III Theory Advanced Python Programming	3	5	3	25	75	100
Core - IV Practical - I Advanced Data Structure and Algorithms using Python Programming Practical	3	5		40	60	100
Elective - I Theory R Programming / Operating Systems Using Unix / Data Visualization	3	5	3	25	75	100
Elective - II Practical - II R Programming Lab Practical/ Operating System Practical using Unix / Data Visualization Practical	3	5	3	40	60	100
	20	30				

Course	Number of Credits	Hours Per Week	Examination Duration (hrs)	Marks		
				I. A	ESE	Total
Semester - II						
Core -V Theory Data Base Management System	4	5	3	25	75	100
Core - VI Theory Exploratory Data Analytics	4	5	3	25	75	100
Core - VII Theory Statistics for Data Science	4	5		25	75	
Core - VIII Practical - III RDBMS with VB .NET Practical	3	4	3	40	60	100
Elective - III Theory Software Engineering and Testing/ Cyber Security / Object Oriented Programming Java	3	5	3	25	75	100
Elective - IV Practical - IV Software Testing Practical/ Cyber Security Practical/ Object Oriented Programming Java Practical	3	4	3	40	60	100
NME - I Fundamentals of Human Rights	2	2	3	25	75	100
Total	23	30				

Course	Number of Credits	Hours Per Week	Examination Duration (hrs)	Marks		
				I. A	ESE	Total
Semester - III						
Core IX Theory Web Technology (React,MangoDB,Node JS)	4	5	3	25	75	100
Core X Theory Advanced Machine Learning	4	5	3	25	75	100
Core XI Theory Data Science and Analytics	5	5	3	25	75	100
Core XII Practical - V Advanced Machine Learning Practical	4	5	3	40	60	100
Elective -V Theory Cloud Computing / Software Development Technology / Social Network analysis.	3	5	3	25	75	100
NME - II Web Technology Lab Practical - VI	2	5	3	40	60	100
Internship Industrial Activity	2	-	-	-	100	100
Total	24	30				

Course	Number of Credits	Hours Per Week	Examination Duration (hrs)	Marks		
				I. A	ESE	Total
Semester - IV						
Core - XIII Theory Data Ethics, privacy and Governance	4	5	-	25	75	100
Elective - VI Natural Language Processing / Computer Vision / Parallel and Distributed Computing	3	5		40	60	100
Core - XIV Project Viva voce	14	16		20	60+20	100
Skill Enhancement Professional Competency Skill	2	4				
Extension Activity	1					
Total	24	30				
Total Credits	91					

Component wise Credit Distribution

Credits	Sem I	Sem II	Sem III	Sem IV	Total
Part A	14	15	17	18	64
Part B	6	6	5	3	20
(i) Discipline– Centric / Generic Skill					
(ii) Soft Skill				2	2
(iii) Summer Internship / Industrial Training			2		2
Part C		2		1	3
Total	20	23	24	24	91

SEMESTER -I

Title of the Paper	Mathematics for Data Science		
Core-I – Theory	I Year & I Semester	Credit:4	

Course Objective:

- To provide a strong foundation in differential and integral calculus concepts.
- To develop analytical skills for solving real-world problems using calculus.
- To understand functions of several variables and their applications in engineering.
- To introduce ordinary differential equations and numerical solution techniques.
- To familiarize students with partial differential equations and their physical applications.

UNIT I : Differential Calculus -Real number system, finite, countable and uncountable sets, sequence and series, relation, function, continuity, differentiation, mean value theorem, Taylor's theorem, economic applications, marginal and elasticity concepts.

UNIT II : Integral Calculus-Riemann integral, properties of integrals, fundamental theorems of calculus, improper integral of first & second kind, comparison test, absolute convergence, applications of definite integral.

UNIT III: Functions of Several Variable - Higher order derivatives and Young's theorem, properties of linear homogeneous functions, Euler's theorem, implicit function theorem, double integral, Fubini's theorem, change of variables in a triple integral, Green's Theorem, Stokes' Theorem, divergence theorem.

UNIT IV: Ordinary Differential Equations - Introduction to differential equation, order of differential equation, first order, Concept of solutions, Geometrical interpretations, Separable form, Reduction to separable form, Exact equations, Integrating factors (of the form $F(x)$ and $F(y)$), linear equations, Bernoulli equation, Picard's existence and uniqueness theorem, Picard's iteration method. numerical methods: Euler's method, improved Euler's method, second order linear ODE, Wronskian, reduction of order, characteristic equations, non-homogeneous equations.

UNIT V: Partial Differential Equations - Classification of PDEs, method of characteristics, local existence results, classification of linear second order PDEs, heat, Laplace and wave equations, conservation laws, Green's function.

TEXTBOOKS:

1. Walter Rudin: Principles of Mathematical Analysis, Third Edition, M.Graw-Hill, Inc.
2. E. A. Coddington: An Introduction to Ordinary Differential Equations, Dover Publications, Inc.
3. Y. Pinchover and J. Rubenstein: An Introduction to Partial Differential Equations.
Robert. C. Mac Owen: Partial Differential Equations, First Edition, Cambridge University Press.
4. L.C. Evans, Partial Differential Equations, Graduate Studies in Mathematics, Volume 19, American Mathematical Society

REFERENCE BOOKS :

1. Erwin Kreyszig: Advanced Engineering Mathematics, Wiley, **2011**.
2. George B. Thomas and Maurice D. Weir: Thomas' Calculus, Pearson, **2018**.

Course Outcome :

CO1	Apply differential and integral calculus concepts to solve computational problems.	K2	IO
CO2	Analyze multivariable functions and optimization techniques.	K4,K5	HO
CO3	Solve ordinary differential equations for mathematical modelling.	K5	HO
CO4	Apply partial differential equations in engineering applications.	K1	LO
CO5	Interpret mathematical models for data science applications.	K5	HO

Course Mapping :

CO/P O	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	-	-	M	-	-	-	-	-	-
CO2	S	S	-	-	-	M	-	-	-	-	-	-
CO3	S	S	-	M	-	M	-	-	-	-	-	-
CO4	S	S	-	M	-	M	-	-	-	-	-	-
CO5	S	S	-	M	M	M	-	-	-	-	-	-

Title of the Paper	Advanced Data Structures and Algorithms		
Core–II – Theory	I Year & I Semester	Credit:4	

Course Objective:

- Define the basic concepts of algorithms and analyze the performance of algorithms.
- Discuss various algorithm design techniques for developing algorithms.
- Discuss various searching, sorting and graph traversal algorithms.
- Understand NP completeness and identify different NP complete Problems.
- Discuss various advanced topics on algorithms.

UNIT I: Introduction: Algorithm, Pseudo code for expressing algorithms, Performance Analysis- Space complexity, Time complexity, Asymptotic Notation- Big oh notation, Omega notation, Theta notation and Little oh notation, Probabilistic analysis, Amortized analysis.

UNIT II: Insertion and deletion and merging with 1) binary search tree, 2) AVL tree, 3) Red Black tree, 4) B tree, 5) B+ tree and Comparison of previous tree structures. Fibonacci Heap, Fibonacci Heap Operations: Find minimum, merge, insert, extract minimum, decrease key and delete, Complexity analysis of the above data structure operations.

UNIT III: Representations of Graphs, Minimum Spanning Trees: Growing a Minimum Spanning Tree – Kruskal and Prim- Single-Source Shortest Paths: The Bellman-Ford algorithm – Single-Source Shortest paths in Directed Acyclic Graphs – Dijkstra ‘s Algorithm, Divide and conquer: General method, applications - Quick sort, Merge sort, Strassen’s matrix multiplication, External Sort: External merge sort, K-Way Merge sorting

UNIT IV: Greedy method: General method, applications-Job sequencing with deadlines, 0/1, knapsack problem, Huffman Codes, Dynamic Programming: General method, applications- Matrix chain multiplication, 0/1 knapsack problem, Traveling salesperson problem, Reliability design.

UNIT V: Backtracking: General method, applications-n-queen problem, sum of subsets problem, graph coloring, Hamiltonian cycles. Branch and Bound: General method, applications - Traveling salesperson problem, 0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution. NP-Hard and NP-Complete problems

Text Books:

1. Peter Brass; Advanced Data Structures; CAMBRIDGE UNIVERSITY PRESS;2008
2. S. Dasgupta, C. Papadimitrou, U Vazirani; Algorithms; Mc Graw Hill;2022
3. J. Klienbergl and E. Tardos, Algorithm Design, Pearson EducationLimited;2013.
4. Ellis Horowitz, Sartaj Sahni, Rajasekharan, Fundamentals of Algorithms, 2nd Edition, Universities Press, 2009.

Reference Books:

- 1.Sartaj Sahni, Data Structures Algorithms and Applications in C++, 2nd Edition, Universities Press, 2007.
2. Aho V Alfred, Hapcroft E John, Ullman D Jeffry, Data Structures and Algorithms, Pearson Education, 2001.
3. Adam Drozdek, Thomson, Data Structures and Algorithms in JAVA, 3rd Edition, Cengage Learning, 2008.
4. Horowitz, Sahni, Mehta, Fundamentals of Data Structures in C++, 2nd Edition, Universities Press, 2007.

Web References:

<https://nptel.ac.in/courses/10602064>

Course Outcome :

CO1	Analyze complexity of algorithms using asymptotic notation.	K2	IO
CO2	Implement advanced data structures for efficient problem solving.	K4,K5	HO
CO3	Design graph and tree-based algorithmic solutions.	K5	HO
CO4	Apply dynamic programming and greedy paradigms.	K1	LO
CO5	Evaluate advanced algorithmic techniques for optimization.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	-	-	M	-	-	-	-	-	-
CO2	S	S	-	M	M	M	-	-	-	-	-	-
CO3	S	S	-	M	M	M	-	-	-	-	-	-
CO4	S	S	-	S	M	M	-	-	-	-	-	-
CO5	S	S	-	S	M	M	-	-	-	-	-	-

Title of the Paper	Advanced Python Programming		
Core–III – Theory	I Year & I Semester	Credit:3	

Course Objectives:

- To acquire programming skills in core Python
- To learn Strings and function
- To develop object oriented skills in Python
- To comprehend various Python Packages
- To develop web applications using Django

UNIT I: Introduction : Fundamental ideas of Computer Science - Strings, Assignment, and Comments - Numeric Data types and Character sets – Expressions – Loops and Selection Statements: Definite iteration: the for Loop - selection: if and if-else statements - Conditional iteration: the while Loop

UNIT II: Strings and Text Files: Accessing Characters and substrings in strings - Data encryption- Strings and Number systems- String methods – Text - Lists and Dictionaries: Lists – Dictionaries – Design with Functions: A Quick review - Problem Solving with top-Down Design - Design with recursive Functions - Managing a Program’s namespace - Higher-Order Functions

UNIT III: Design with Classes: Getting inside Objects and Classes – Data-Modeling Examples – Building a New Data Structure – The Two – Dimensional Grid - Structuring Classes with Inheritance and Polymorphism-Graphical User Interfaces-The Behavior of terminal-Based programs and GUI-Based programs - Coding Simple GUI-Based programs - Windows and Window Components - Command Buttons and responding to events

UNIT IV: Working with Python Packages: NumPy Library-Ndarray – Basic Operations – Indexing, Slicing and Iteration – Array manipulation - Pandas –The Series – The DataFrame - The Index Objects – Data Vizualization with Matplotlib – The Matplotlib Architecture – pyplot – The Plotting Window – Adding Elements to the Chart – Line Charts – Bar Charts – Pie charts.

UNIT V: Django: Installing Django – Building an Application – Project Creation – Designing the Data Schema - Creating an administration site for models - Working with QuerySets and Managers – Retrieving Objects – Building List and Detail Views.

Text Book:

1. K.A. Lambert, “ Fundamentals of Python: first programs”, Second Edition, Cengage Learning, 2018 (Unit - I, II and III)
2. Fabio Nelli, “Python Data Analytics: With Pandas, NumPy, and Matplotlib”, Second Edition, Kindle Edition, 2018 (Unit - IV)
3. Antonio Mele, “Django 3 By Example”, Third Edition, 2020 (Unit - V)

Course Outcome :

CO1	Develop Python programs using control structures and functions.	K2	IO
CO2	Apply object-oriented programming concepts in Python.	K4,K5	HO
CO3	Use Python libraries for data analysis and visualization.	K5	HO
CO4	Build database/web-enabled Python applications.	K1	LO
CO5	Design modular and reusable Python solutions.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	R programming		
Elective-I Theory	I Year & II Semester	Credit:3	

Course Objectives:

- The basics of statistical computing and data analysis.
- How to use R for analytical programming.
- How to implement data structure in R?
- R loop functions and debugging tools.

UNIT I: Introduction to Data Analysis Overview of Data Analytics, Need of Data Analytics, Nature of Data, Classification of Data: Structured, Semi-Structured, Unstructured, Characteristics of Data, Applications of Data Analytics.

UNIT II: R Programming Basics Overview of R programming, Environment setup with R Studio, R Commands, Variables and Data Types, Control Structures, Array, Matrix, Vectors, Factors, Functions, R packages.

UNIT III: Data Visualization using R Reading and getting data into R (External Data): Using CSV files, XML files, Web Data, JSON files, Databases, Excel files. Working with R Charts and Graphs: Histograms, Boxplots, Bar Charts, Line Graphs, Scatter plots, Pie Charts

UNIT IV: Statistics with R Random Forest, Decision Tree, Normal and Binomial distributions, Time Series Analysis, Linear and Multiple Regression, Logistic Regression, Survival Analysis

UNIT V: Prescriptive Analytics Creating data for analytics through designed experiments, Creating data for analytics through active learning, Creating data for analytics through reinforcement learning.

Text Book:

1. An Introduction to R, Notes on R: A Programming Environment for Data Analysis and Graphics. W. N. Venables, D.M. Smith and the R Development Core Team. URL: <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf> 5.

Reference Books:

1. Jared P Lander, R for everyone: advanced analytics and graphics, Pearson Education, 2013.
2. Dunlop, Dorothy D., and Ajit C. Tamhane. Statistics and data analysis: from elementary to intermediate. Prentice Hall, 2000.
3. G Casella and R.L. Berger, Statistical Inference, Thomson Learning 2002.
4. P. Dalgaard, Introductory Statistics with R, 2nd Edition. (Springer 2008).
5. Michael Berthold, David J. Hand, Intelligent Data Analysis, Springer.
6. Hastie, Trevor, et al. The elements of statistical learning. Vol. 2. No. 1. New York: springer, 2009.
7. Montgomery, Douglas C., and George C. Runger. Applied statistics and probability for engineers. John Wiley & Sons, 2010.

Course Outcome :

CO1	Explain R programming environment and syntax.	K2	IO
CO2	Manipulate data using R data structures and functions.	K4,K5	HO
CO3	Perform statistical analysis using R packages.	K5	HO
CO4	Create visualizations and reports using R.	K1	LO
CO5	Develop analytical solutions using R for data science tasks.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	R programming Lab		
Elective - I - Practical - I	I Year & II Semester	Credit: 3	

Course Objectives:

- The basics of statistical computing and data analysis.
- How to use R for analytical programming.
- How to implement data structure in R.
- R loop functions and debugging tools.

List of Experiments:

1. R AS CALCULATOR APPLICATION

- Using with and without R objects on console
- Using mathematical functions on console
- Write an R script, to create R objects for calculator application and save in a specified location in disk.

2. DESCRIPTIVE STATISTICS IN R

- Write an R script to find basic descriptive statistics using summary, str, quartile function on mtcars & cars datasets.
- Write an R script to find subset of dataset by using subset (), aggregate () functions on iris dataset.

3. READING AND WRITING DIFFERENT TYPES OF DATASETS

- Reading different types of data sets (.txt, .csv) from Web and disk and writing in file in specific disk location.
- Reading Excel data sheet in R.
- Reading XML dataset in R

4. VISUALIZATIONS

- Find the data distributions using box and scatter plot.
- Find the outliers using plot.
- Plot the histogram, bar chart and pie chart on sample data.

5. CORRELATION AND COVARIANCE

- a. Find the correlation matrix.
- b. Plot the correlation plot on dataset and visualize giving an overview of relationships among data on iris data.
- c. Analysis of covariance: variance (ANOVA), if data have categorical variables on iris data

6. REGRESSION MODEL

Import a data from web storage. Name the dataset and now do Logistic Regression to find out relation between variables that are affecting the admission of a student in a institute based on his or her GRE score, GPA obtained and rank of the student. Also check the model is fit or not. Require (foreign), require (MASS).

7. MULTIPLE REGRESSION MODEL

Apply multiple regressions, if data have a continuous Independent variable.
Apply on above dataset.

8. REGRESSION MODEL FOR PREDICTION

Apply regression Model techniques to predict the data on above dataset.

9. CLASSIFICATION MODEL

- a. Install relevant package for classification.
- b. Choose classifier for classification problem.
- c. Evaluate the performance of classifier.

10. CLUSTERING MODEL

- a. Clustering algorithms for unsupervised classification.
- b. Plot the cluster data using R visualizations.

Course Outcome :

CO1	Develop programs using R syntax and data types.	K2	IO
CO2	Perform data manipulation and statistical analysis in R.	K4,K5	HO
CO3	Create visualizations using R packages.	K5	HO
CO4	Apply R for predictive modelling tasks.	K1	LO
CO5	Build reproducible analytical workflows in R.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	M	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	S	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	Operating Systems and Unix		
Elective - II Theory	I Year & I Semester	Credit: 3	

Course Objectives:

- To understand the basic concepts and functions of operating systems
- To understand Processes and Threads
- To analyze Scheduling algorithms.
- To understand the concept of Deadlocks.
- To analyze various memory management schemes.
- To understand I/O management and File systems.

UNIT I: Introduction- Operating Systems- Computer System Architecture- Operating System Structure- Operating System Services- User and Operating System Interface- System Calls- Types of System Calls- System Programs- Operating System Debugging –Operating System Generation- System Boot-Process - Process Scheduling- Interprocess Communication- Examples of IPC Systems- Communication in Client– Server Systems.

UNIT II: Threads - Multicore Programming- Multithreading Model- Threading Issues- Process Synchronization- The Critical-Section Problem- Peterson’s Solution- Synchronization Hardware - Mutex Locks- Semaphores- Monitors- CPU Scheduling- Scheduling Criteria- Scheduling Algorithm- Thread Scheduling- Multiple Processor Scheduling- System Model- Methods for Handling Deadlocks- Deadlock Prevention- Deadlock Avoidance- Deadlock Detection.

UNIT III: Main Memory- Swapping- Contiguous Memory Allocation- Segmentation- Paging- Structure of the Page Table- Example: ARM Architecture- Virtual Memory- Demand Paging- Page Replacement- Thrashing- Memory-Mapped Files- Allocating Kernel Memory

UNIT IV: Mass-Storage Structure- Disk Structure-Disk Attachment-Disk Scheduling -Disk Management –Swap Space Management RAID Structure – Stable Storage Implementation- File Concept -Access Methods -Directory and Disk Structure - File-System Mounting - File Sharing – Protection- File-System Structure File System Implementation-Directory Implementation-

Allocation Methods - Free Space Management Efficiency and Performance - Recovery – NFS- I/O Hardware -Application I/O Interface- Kernel I/O Subsystem - Transforming I/O Requests to Hardware Operations.

UNIT V: Protection- Goals of Protection-Principles of Protection-Domain of Protection-Access Matrix-

Implementation of the Access Matrix-Access Control -Revocation of Access Rights- Capability-Based Systems-Language Based Protection- The Security Problem-Program Threats- System and Network Threats -Cryptography as a Security Tool -User Authentication-Implementing Security Defenses-Firewalling to Protect Systems and Networks -Computer Security Classifications

TEXT BOOK:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, - Operating System Concepts, 9th Edition, John Wiley and Sons Inc., 2012.

REFERENCES:

1. RamazElmasri, A. Gil Carrick, David Levine, —Operating Systems – A Spiral Approach, Tata McGraw Hill Edition, 2010
2. AchyutS.Godbole, AtulKahate, —Operating Systems, McGraw Hill Education, 2016.
3. Andrew S. Tanenbaum, —Modern Operating Systems, Second Edition, Pearson Education, 2004.

E-Resources:

<https://applied-programming.github.io/Operating-Systems-Notes/>

<https://ecomputernotes.com/fundamental/disk-operating-system/what-is-operating-system>

Course Outcome :

CO1	Explain UNIX operating system concepts and architecture.	K2	IO
CO2	Use UNIX commands and shell scripting effectively.	K4,K5	HO
CO3	Manage processes, memory, and file systems in UNIX.	K5	HO
CO4	Apply inter-process communication and scheduling concepts.	K1	LO
CO5	Administer UNIX systems for development environments.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Operating System and unix lab		
Elective - II Practical - II	I Year & I Semester	Credit: 3	

Course Objectives

- To enable the students to study and understand the efficiency of Linux shell script.
- To demonstrate the File Backup process.
- To develop and implement the shell script for GUI processing.
- To develop and implement the shell script for IPC and Networking.
- To demonstrate PostgreSQL.

List of Programs

1. File management system calls: Write a program to implement
 1. create a file
 2. Copy one file to another
 3. Linking a file
 4. Delete a file.
2. Directory management system calls: Write a program to change directory and print its contents.
3. Parent process – Child process Relationship.
4. Implementing IPC using pipes.
5. Simulation of scheduling algorithms: Write a program to implement the following process scheduling algorithms
 - 1) First Come First Serve
 - 2) Shortest Remaining Job First
 - 3) Round Robin

- 4) Preemptive Priority Scheduling
6. Implementation of semaphore: Write a program that demonstrates how two processes can share a variable using semaphore.
7. Implementation of shell : Write a 'c' program to implement a shell.
8. Producer – Consumer Problem :Write a program to implement producer consumer problem (Using POSIX semaphores)
9. To implement Banker's algorithm for a multiple resources.
10. To study page replacement policies like
 - 1) OPTIMAL
 - 2) LEAST RECENTLY USED(LRU)
 - 3) FIRST-IN-FIRST-OUT

Course Outcome :

CO1	Demonstrate UNIX commands and shell usage.	K2	IO
CO2	Develop shell scripts for automation.	K4,K5	HO
CO3	Implement process and file management operations.	K5	HO
CO4	Apply IPC and scheduling concepts practically.	K1	LO
CO5	Debug and optimize UNIX-based programs.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	M	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	DATA VISUALIZATION		
Elective	IYear- II Semester	Credit:5	

COURSE OBJECTIVES:

- To understand the fundamentals of data visualization.
- To know the working principles of various information visualization depth tools.
- To acquire knowledge about the issues in data representation.
- To visualize the Data using tools Tableau
- To gain skill in designing real time interactive information visualization system.

UNIT1: Introduction-Context of data visualization – Definition, Methodology, Visualization design objectives.Key Factors–Purpose,visualization function and tone,visualization design options–Data representation,Data Presentation,Seven stages of data visualization,widgets,data visualization tools. Mapping - Time Series - Connections and Correlations – Scatter plot Maps-Trees,Hierarchies,and Recursion-Networks and Graphs

UNITII :Visualization techniques for time-series, trees & graphs Mapping -Times eries- Connections and correlations–Indicator -Area chart- Pivottable Scatter charts,Scatter maps- Tree maps,Spacefillingandnon-spacefillingmethodsHierarchies and Recursion - Networks and Graphs- Displaying Arbitrary Graphs-node link graph-Matrix representation for graphs -Infographics

UNIT III: Text and document visualization - Acquiring data-Where to Find Data, Tools for Acquiring Data from the Internet,Locating Files for Use with Processing, Loading Text Data, Dealing with Files and Folders, Listing Files in a Folder,Asynchronous Image Downloads,Web Techniques,Parsing data Levels of Effort,Tools for Gathering Clues,Text Markup Languages,Regular Expressions,Grammars and BNF Notation, Compressed Data, Vectors and Geometry, Binary Data Formats, Advanced Detective Work.

UNIT IV: Interactive data visualization - Drawing with data – Scales – Axes – Updates, Transition and Motion – Interactivity-Layouts–Geomapping–Exporting,Framework–D3.js,Tableau Dashboards

UNIT V: Security in data visualization Port scan visualization-Vulnerability assessment and exploitation - Firewalling visualization - Intrusion detection log visualization -Attacking and defending visualization systems–Creating secured visualization system.

TEXTBOOKS

- 1.Robert Spence, “Information Visualization An Introduction”, Third Edition, Pearson Education, 2014.
2. Colin Ware, “Information Visualization Perception for Design”, Third edition, Morgan Kaufmann Publishers, 2012.
- 3.Robert Spence, “ Information Visualization Design for Interaction”, Second Edition, Pearson Education, 2006.
- 4.Benjamin B. Bederson and Ben shneiderman, “The Craft of Information Visualization”, Morgan Kaufmann Publishers, 2003.
- 5.Thomas strothotte, “Computational Visualization: Graphics, Abstraction and Interactivity”, Springer, 1998.
- 6.Matthew O. Ward, George Grinstein, Daniel Keim, “Interactive Data
- 7.Visualization: Foundation, Techniques and Applications”, Second Edition, A.K. Peters/CRC Press, 2015. 7. Joerg Osarek, “Virtual Reality Analytics”, Gordon’s Arcade, 2016.

REFERENCES

- 1.Benjamin B.Bederson and Bensneider man,“The Craft of Information Visualization”, Morgan Kaufmann Publishers,2003.
- 2.Thomas strothotte,“Computational Visualization:Graphics,Abstraction and Interactivity”, Springer,1998.
- 3.Matthew O. Ward, George Grinstein, Daniel Keim, “Interactive Data Visualization:Foundation,Techniques and Applications”,Second Edition,K.Peters /CRCPress ,2015.
- 4.JoergOsarek,“VirtualRealityAnalytics”,Gordon’sArcade,2016.

Course Outcome :

CO1	Explain principles of effective data visualization and visual perception.	K2	IO
CO2	Create charts, dashboards, and reports using visualization tools.	K4,K5	HO
CO3	Apply visualization techniques for multidimensional datasets.	K5	HO
CO4	Interpret visual analytics for business and decision-making contexts.	K1	LO
CO5	Design interactive and story-driven visual reports.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	S	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	M	S	S	S	-	-	-	-	-	-

Title of the Paper	DATA VISUALIZATION LAB		
Elective - I	I Year- I Semester	Credit:5	

COURSE OBJECTIVES:

- Understand the importance of data visualization for business intelligence and decision making.
- Know approaches to understand visual perception
- Learn about categories of visualization and application areas
- Familiarize with the data visualization tools
- Gain knowledge of effective data visuals to solve workplace problems

List of Experiments :

1. Introduction to various Data Visualization tools
2. Basic Visualization in Python
3. Basic Visualization in R
4. Introduction to Tableau and Installation
5. Connecting to Data and preparing data for visualization in Tableau
6. Data Aggregation and Statistical functions in Tableau
7. Data Visualizations in Tableau
8. Basic Dashboards in Tableau

Course Outcome :

CO1	Explain principles of effective data visualization.	K2	IO
CO2	Create charts and dashboards using visualization tools.	K4,K5	HO
CO3	Apply visualization techniques to large datasets.	K5	HO
CO4	Interpret visual analytics for decision-making.	K1	LO
CO5	Design interactive visual reports.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	S	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	Advanced Data Structures and Algorithms Practical		
Core–IV - Practical	I Year & I Semester	Credit:3	

Course Objectives:

- To provide the foundations of the practical implementation and usage of Algorithms and Data Structures.
- To ensure that the student evolves into a competent programmer capable of designing and analysing implementations of algorithms and data structures for different kinds of problems.
- To expose the student to the algorithm analysis techniques,
- To make the students understand the theory of reductions, and to the classification of problems.

List of Experiments:

Implement the following using Java or Python

1. Write a program to perform the following operations on a heterogeneous singly linked list.
i) Creation ii) Insertion iii) Deletion iv) Traversal.
2. Write a program to perform the following operations on a heterogeneous doubly linked list.
i) Creation ii) Insertion iii) Deletion iv) Traversal in both ways
3. Write a program that implements using java generic class, the stack (its operations)
4. Write a program that implements using java generic class, the queue (its operations)
5. Write a program that implements the Quick sort method.
6. Write a program that implement the Merge sort method.
7. Write a program that implement the SHELL sort method.
8. Write a program to perform the following:
i) Creating a Binary Tree of integers ii) Traversing the above binary tree in preorder, inorder and postorder.
9. Write a program to perform the following:
i) Creating a AVL Tree ii) insertion iii) deletion iv) Traversing the above AVL tree in preorder, inorder and postorder.
10. Write a program that uses functions to perform the following:
i) Creating a SplayTree ii) traverse

11. Write a program to perform the following:

i) Creating a B-Tree of integers ii) insertion iii) deletion

12. Write a program that implements Kruskal's algorithm using a disjoint set data structure. The program takes as input a file (data.txt), in which each line either represents a vertex or an edge. For the edge lines, the first integer on that line representing the starting vertex, the second the ending vertex, and the third the weight of the edge. Use this file to construct, line by line, the graph upon which Kruskal's algorithm will be run (do NOT hardcode this graph!).

13. Write a program to simulate various graph traversing algorithms.

14. Write a program to find the minimal spanning tree of a graph using the Prim's algorithm.

15. Write a program to find shortest path using Bellman Ford's Algorithm

Course Outcome :

CO1	Implement advanced linear and non-linear data structures in programming environments.	K2	IO
CO2	Develop programs for tree and graph traversal/processing algorithms.	K4,K5	HO
CO3	Apply searching and sorting techniques to solve computational problems.	K5	HO
CO4	Analyze and compare algorithm performance through practical implementation.	K1	LO
CO5	Debug, optimize, and document algorithmic solutions effectively.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	M	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	M	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

SEMESTER -II

Title of the Paper	Database Management Systems		
Core -V	I Year & II Semester	Credit: 4	

Course Objective:

- To learn the fundamentals of data models and to represent a database system using ERdiagrams.
- To study SQL and relational database design.
- To understand the internal storage structures using different file and indexing techniques which will help in physical DB design.
- To understand the fundamental concepts of transaction processing- concurrency control techniques and recovery procedures.
- To have an introductory knowledge about the Storage and Query processing Techniques

UNIT I: Introduction to Database Systems – Relational Model – Structure – Relational Algebra – Null Values – SQL – Set Operation – Views – Advanced SQL – Embedded SQL – Recursive Queries – The Tuple Relational Calculus – Domain Relational Calculus.

UNIT II: E-R Model – Constraints – E-R- Diagrams Weak Entity Sets – Reduction to Relational Schemes – Relational Database Design – Features of Relational Design – Automatic Domains and First Normal Form – Decomposition using Functional Dependencies – Multivalued Dependencies – More Normal Forms – Web Interface – Object – Based Databases – Structured Types and inheritance in SQL – Table inheritance – Persistent.

UNIT III: Storage and File Structure – RAID – File Organization – Indexing and Hashing – B Tree – B Tree Index files - Static and Dynamic Hashing – Query Processing – Sorting & Join Operators – Query Optimization – Choice of Evaluation Plans.

UNIT IV: Transaction Management – Implementation of Atomicity and Durability – Serializability – Recoverability – Concurrency Control – Dead Lock Handling – Recovery System – Buffer Management.

UNIT V: Database – System Architecture – Client Server – Architectures – Parallel System – Network Types – Distributed Database – Homogeneous and Hetrogeneous Database – Directory System – Case Study – Oracle – MSSQL Server.

Recommended Text

1. A. Silberschatz, H.F. Korth and S. Sudharshan, 2006, Database System Concepts, 5th Edition, Tata McGraw Hill, New Delhi.

Reference Books

1. J. D. Ullman, 1988, Principles of Database Systems, Galgotia Publishers, New Delhi.
2. C.J. Date, 1985, An Introduction to Database Systems, Third Edition, Narosa, New Delhi.
3. Elmasri and Navathe, 1999, Fundamentals of Database Systems, Third Edition, Pearson Education, Delhi.
4. C. Ritchie, 2004, Relational Database Principals, 2nd Edition, Thomson, Singapore.

Website and e-Learning Source :

- 1) <http://www.cse.iitb.ac.in/dbms/DataPapers-Local/DBConceptsBook/slide-dir/>
- 2) <http://www.cse.iitb.ac.in/dbms/DataPapers-Local/DBConceptsBook/slide-dir/>

Course Outcome :

CO1	Design relational database models using ER concepts.	K2	IO
CO2	Construct SQL queries for database operations.	K4,K5	HO
CO3	Normalize relational schemas for efficient storage.	K5	HO
CO4	Apply transaction and concurrency mechanisms.	K1	LO
CO5	Optimize database performance using indexing/query techniques.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	M	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	M	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	EXPLORATORY DATA ANALYSIS		
Core -VI	IIYear- II Semester	Credit:5	

Course Objectives:

- To outline an overview of exploratory data analysis.
- To implement data visualization using Matplotlib.
- To perform univariate data exploration and analysis.
- To apply bivariate data exploration and analysis.
- To use Data exploration and visualization techniques for multivariate and time series data.

UNIT I: Exploratory Data Analysis- EDA fundamentals – Understanding data science - Significance of EDA – Making sense of data – Comparing EDA with classical and Bayesian analysis – Software tools for EDA – Visual Aids for EDA – Data transformation techniques-merging database, reshaping and pivoting, Transformation techniques.

UNITII: Eda Using Python - Data Manipulation using Pandas – Pandas Objects – Data Indexing and Selection – Operating on Data – Handling Missing Data – Hierarchical Indexing – Combining datasets – Concat, Append,Merge and Join–Aggregation and grouping – Pivot Tables – Vectorized String Operations.

UNIT III: Univariate analysis Introduction to Single Variable :Distribution Variables – Numerical Summaries of Level and Spread–Scaling and Standardizing–Inequality.

UNIT IV: Bivariate analysis -Relationships between Two Variables – Percentage Tables – Analysis Contingency Tables –Handling Several Batches–Scatter plots and Resistant Lines.

UNIT V: Multivariate andtime series analysis Introducing a Third Variable – Causal Explanations – Three-Variable Contingency Tables and Beyond – Fundamentals of TSA – Characteristics of time series data – Data Cleaning – Time-based indexing–Visualizing–Grouping–Resampling.

TEXTBOOKS

1. Suresh Kumar Mukhiya, UsmanAhmed,“Hands-On Exploratory Data Analysis with Python”,PacktPublishing,2020.(Unit1)
2. Jake Vander Plas,“Python DataScience Handbook: Essential Tools for Working with Data”. First Edition,O Reilly,2017.(Unit2)
3. CatherineMars,JaneElliott,“ExploringData:An Introduction to Data Analysis for Social Scientists”,WileyPublications,2ndEdition,2008.(Unit3,4,5)

REFERENCES

1. Eric Pimpler, Data Visualization and Exploration with R, Geo Spatial Training service, 2017.
2. Claus O. Wilke, "Fundamentals of Data Visualization", O'Reilly Publications, 2019.
3. Matthew O. Ward, Georges Grinstein, Daniel Keim, "Interactive Data Visualization: Foundations, Techniques, and Applications", 2nd Edition, CRC Press, 2015.

Course Outcome :

CO1	Explain EDA concepts and preprocessing methods.	K2	IO
CO2	Perform data cleaning and transformation techniques.	K4, K5	HO
CO3	Conduct univariate and bivariate analysis.	K5	HO
CO4	Analyze multivariate datasets statistically.	K1	LO
CO5	Interpret insights from exploratory analysis for decision making.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	RDBMS LAB with VB.NET		
Core - VII	IIYear- II Semester	Credit:5	

OBJECTIVES:

- To understand data definitions and data manipulation commands
- · To learn the use of nested and join queries
- · To understand functions, procedures and procedural extensions of databases
- · To be familiar with the use of a front end tool
- · To understand design and implementation of typical database applications

List of Experiments:

1. Data Definition Commands, Data Manipulation Commands for inserting, deleting, updating and retrieving Tables and Transaction Control statements.
2. Database Querying–Simple queries, Nested queries, Sub queries and Joins
3. Views, Sequences, Synonyms
4. Database Programming: Implicit and Explicit Cursors
5. Procedures and Functions
6. Triggers
7. Exception Handling
8. Database Design using ER modeling, normalization and Implementation for any
9. application
10. Database Connectivity with Front End Tools
11. Case Study using real life database applications.
 - a) Library Information Processing.
 - b) Students Mark sheet processing using images.
 - c) Bank Transactions (SB).
 - d) Pay roll processing.
 - e) Inventory
 - f) Purchase order processing.

Course Outcome :

CO1	Design database-driven VB.Net applications.	K2	IO
CO2	Connect applications to relational databases.	K4,K5	HO
CO3	Implement CRUD operations using SQL and VB.Net.	K5	HO
CO4	Develop form-based data entry/reporting systems.	K1	LO
CO5	Debug and deploy database applications.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	M	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Cyber Security		
Elective - III Theory	I Year & II Semester	Credit: 3	

Course Objectives:

- To understand the basics of Cybercrime and Computer forensics with protecting mechanism
- To explore the working principles of WLAN, Email and Smartphone along with security mechanism and guidelines
- To gain the ability to understand the importance of cyber investigations with its functioning role and learn the basics of Wi Fi and its security measures
- To understand and learn the method of seize the digital evidence
- To learn and analyze the concepts of digital forensics with cybercrime prevention techniques

UNIT I: Introduction to cybercrime: Classification of cybercrimes – reasons for commission of cybercrime – malware and its type – kinds of cybercrime – authentication – encryption – digital signatures – antivirus – firewall – steganography – computer forensics – why should we report cybercrime – introduction counter cyber security initiatives in India – generating secure password – using password manager-enabling two-step verification – security computer using free antivirus.

UNIT II: Tips for buying online: Clearing cache for browsers – wireless LAN-major issues with WLAN-safe browsing guidelines for social networking sites – email security tips – introduction-smartphone security guidelines – purses, wallets, smart phones – platforms, setup and installation-communicating securely with a smartphone.

UNIT III: Cyber investigation roles: Introduction – role as a cybercrime investigator – the role of law enforcement officers – the role of the prosecuting attorney – incident response: introduction-post mortem versus live forensics – computer analysis for the hacker defender program-network analysis – legal issues of intercepting Wi-Fi transmission – Wi-Fi technology – Wi-Fi RF-scanning RF – eavesdropping on Wi-Fi – fourth amendment expectation of privacy in WLAN.

UNIT IV: Seizure of digital information: introduction – defining digital evidence – digital evidence seizure methodology – factors limiting the wholesale seizure of hardware – other options for seizing digital evidence – common threads within digital evidence seizure – determining the most appropriate seizure method– conducting cyber investigations–demystifying computer/cyber crime – IP addresses – the explosion of networking – interpersonal communication.

UNIT V: Digital forensics and analyzing data: introduction – the evolution of computer forensics– phases of digital forensics-collection – examination-analysis – reporting – Cyber crime prevention: Introduction – crime targeted at a government agency.

Text books:

1. Dr.JeetendraPande, “Introduction to Cyber Security” Published by Uttarakhand Open University, 2017.(Chapter: 1.2-6.4,9.3-12.2)
2. Anthony reyes, Kevin o’shea, Jim steele, Jon R. Hansen, Captain Benjamin R. Jean Thomas Ralph, “Cyber-crime investigations” - bridging the gaps between security professionals, law enforcement, and prosecutors, 2007.(Chapter: 4, 5, 6, 7, 8, 9,10)

Reference Books:

1. Sebastian Klipper,“ Cyber Security: Ein Einblick fur Wirtschaftswissenschaftler Fachmedien Wiesbaden,2015.
2. John G.Voller Black and Veatch, “Cyber Security” Published by John Wiley & Sons, Inc., Hoboken, New Jersey Published simultaneously in Canada ©2014.

Course Outcomes

On the successful completion of the course, students will be able to

CO1	Understand, describe, analyze and examine the basics of Cyber security concepts and its implementation in India.	K1-K6
CO2	Comprehend and demonstrate the security tips in browsers, WLAN, social networks, Email security and Smart phone. Apply the investigations in post mortem and Forensics.	K1-K6
CO3	Understand, apply and evaluate the various investigation roles and Wi Fi protecting mechanisms.	K1-K6
CO4	Understand, illustrate and evaluate the method of seize the digital information and evidences forensics data and evaluate the forensics reports.	K1-K6
CO5	Comprehend, apply and appraise the methods digital forensics with cybercrime prevention techniques.	K1-K6

K1- Remember, K2- Understand, K3- Apply , K4- Analyze, K5- Evaluate, K6- Create

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	CO1	S	L	-	L	M	L	M	M	-	M	-
CO2	CO2	M	S	-	L	M	L	M	M	-	M	-
CO3	CO3	M	S	L	L	M	L	M	M	-	M	M
CO4	CO4	S	M	L	S	M	L	S	M	-	M	-
CO5	CO5	M	S	M	L	S	L	M	S	-	S	-

S- Strong; M-Medium; L-Low

Title of the Paper	Cyber Security Lab		
Elective - IV Practical	I Year & II Semester	Credit: 3	

COURSE OBJECTIVES

- To learn and implement to Change the wireless device mode as monitor mode
- To develop in multiple vulnerabilities webserver
- To understand and implement the open ports in the network
- To acquire programming skills in Implement various wireless device modes
- To comprehend related to find the sub domains of webpage

Implement the following using any cyber security tools

1. Install virtual box (kali Linux)
2. Generate a secure password using keepass
3. Change the wireless device mode as monitor mode
4. Find the known and open vulnerabilities of system using metasploit
5. Identify the multiple vulnerabilities webserver using nikto tool
6. Identify the open ports in the network using nmap tools
7. List all the network around us and display the information about the networks
8. Sniff and capture the packet sent over HTTP requests
9. Find the owners of internet resources using Whois Lookup tool
10. Find the subdomains of webpage using knock tool

Course Outcome :

CO1	Demonstrate cybersecurity fundamentals and tools.	K2	IO
CO2	Perform vulnerability assessment and basic hardening.	K4,K5	HO
CO3	Configure authentication and access controls.	K5	HO
CO4	Analyze common attacks and mitigation strategies.	K1	LO
CO5	Apply secure practices in system/network environments.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	M	M	-	S	-	-	-	S	-	-
CO2	S	S	M	M	M	S	-	-	-	S	-	-
CO3	S	S	M	M	M	S	-	-	-	S	-	-
CO4	S	S	M	S	M	S	-	-	-	S	-	-
CO5	S	S	M	S	M	S	-	-	-	S	-	-

Title of the Paper	Software Engineering and Testing		
Elective-III Theory	I Year & II Semester	Credit: 3	

Course Objectives:

- To understand the concepts of analysis, design, and implementation of a software product
- To have general understanding about object oriented software engineering
- To make students to get experience and be ready for the large scale projects in IT industry

Unit I :Introduction:- Evolution – From an Art form on Engineering Discipline: Evolution of an Art into an Engineering Discipline. – Software Development of Projects: Program versus Product – Emergence of Software Engineering: Early Computer Programming – High Level Language Programming – Control Flow-based Design – Data Structure Oriented Design – Object Oriented Design. **Software Life Cycle Models:-** A few Basic Concepts – Waterfall Model and its Extension: Classical Waterfall Model – Iterative Waterfall Model – Prototyping Model – Evolutionary Model. – Rapid Application Development (RAD): Working of RAD. – Spiral Model. (12L)

Unit II Software Project Management:- Responsibilities of a Software Project Manager – Project Planning- Project Estimation Techniques-Risk Management. **Requirements Analysis and Specification:-** Requirements Gathering and Analysis – Software Requirements Specifications (SRS):Users of SRS Document – Characteristics of a Good SRS Document – Important Categories of Customer Requirements – Functional Requirements – How to Identify the Functional Requirements? – Organization of the SRS Document. (12L)

Unit III :Software Design:- Overview of the Design Process: Outcome of the Design Process – Classification of Design Activities. – How to Characterize a good Software Design? **Function-Oriented Software Design:-** Overview of SA/SD Methodology – Structured Analysis – Developing the DFD Model of a System: Context Diagram – Structured Design – Detailed Design. (12L)

Unit IV : User Interface Design:- Characteristics of a good User Interface - Basic Concepts – Types of User Interfaces – Fundamentals of Components based GUI Development: Window System. **Coding and Testing:-** Coding – Software Documentation – Testing: Basic Concepts and Terminologies – Testing Activities. – Unit Testing – Black-box Testing: Equivalence Class Partitioning – Boundary Value Analysis. – White-box Testing. (12L)

Unit V : Software Reliability and Quality Management:- Software Reliability: Hardware versus Software Reliability. – Software Quality – Software Quality Management System –ISO 9000: What is ISO 9000 Certification? – ISO 9000 for Software Industry – Shortcomings of ISO 9000 Certification. – SEI Capability Maturity Model: Level 1 to Level 5. **Software Maintenance:-** Characteristics of Software Maintenance: Characteristics of Software Evolution – Software Reverse Engineering. (12L)

Text Book:

Fundamentals of Software Engineering Fourth Edition by Rajib Mall – PHI Learning Private Limited 2015

Course Outcome :

CO1	Explain software development life cycle models and software process frameworks.	K2	IO
CO2	Apply software requirements analysis and design methodologies.	K4,K5	HO
CO3	Develop software design models using UML and related techniques.	K5	HO
CO4	Prepare and execute software testing strategies and test cases.	K1	LO
CO5	Use software quality assurance and testing tools for validation and verification.	K5	HO

Course Mapping :

CO/P O	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	M	M	-	S	-	-	-	-	-	-
CO2	S	S	M	M	-	S	-	-	-	-	-	-
CO3	S	S	M	S	M	S	-	-	-	-	-	-
CO4	S	S	M	S	M	S	-	-	-	-	-	-
CO5	S	S	M	S	M	S	-	-	-	-	-	-

Title of the Paper	Software Engineering and Testing Lab		
Elective - IV Practical	I Year & II Semester	Credit: 3	

List of Experiments

- 1 Hands on Software Engineering principles Infrastructure.
2. usage of Front-end and Back-end technologies and packages. Prepare the following documents for three of the experiments listed below using software engineering methodology.
3. Program Analysis and Project Planning.
2. Thorough study of the problem – Identify project scope, Objectives,
3. Software requirement Analysis
- 3 Describe the individual Phases / Modules of the project, Identify deliverables
- 4 Software Design
 - a. Use work products – Data dictionary, Use case diagrams and activity diagrams,
build and test class diagrams,
 - b. Sequence diagrams and add interface to class diagrams, DFD, ER diagrams
 - c. Software Development and Debugging using any Front end and Back end tool
 - d. Software Verification and Validation procedures

Course Outcome :

CO1	Apply software testing principles in practical testing scenarios.	K2	IO
CO2	Design and execute unit, integration, and system test cases.	K4,K5	HO
CO3	Use automated testing tools/frameworks for validation.	K5	HO
CO4	Analyze defects and generate comprehensive test reports.	K1	LO
CO5	Validate software quality against specified requirements.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	M	M	-	S	-	-	-	-	-	-
CO2	S	S	M	M	-	S	-	-	-	-	-	-
CO3	S	S	M	S	M	S	-	-	-	-	-	-
CO4	S	S	M	S	M	S	-	-	-	-	-	-
CO5	S	S	M	S	M	S	-	-	-	-	-	-

Title of the Paper	Object Oriented Programming through Java		
Elective - III Theory	I Year & II Semester	Credit: 3	

Course Objectives:



- The objective of this course is to provide object oriented concepts through which robust, secured and reusable software can be developed.
- To understand object oriented principles like abstraction, encapsulation, inheritance,
- polymorphism and apply them in solving problems.
- To understand the principles of inheritance and polymorphism and demonstrate how they relate to the design of abstract classes.
- To understand the implementation of packages and interfaces.
- To understand the concepts of exception handling, multithreading and collection classes.
- To understand how to connect to the database using JDBC.
- To understand the design of Graphical User Interface using applets and swing controls.

UNIT-I :Java Programming- History of Java, comments, Data types, Variables, Constants, Scope and Lifetime of variables, Operators, Type conversion and casting, Enumerated types, Control flow- block scope, conditional statements, loops, break and continue statements, arrays, simple java stand alone programs, class, object, and its methods constructors, methods, static fields and methods, access control, this reference, overloading constructors, recursion, exploring string class, garbage collection.

UNIT – II :Inheritance – Inheritance types, super keyword, preventing inheritance: final classes and methods.. Polymorphism – method overloading and method overriding, abstract classes and methods. Interfaces- Interfaces Vs Abstract classes, defining an interface, implement interfaces, accessing implementations through interface references, extending interface, inner class. Packages- Defining, creating and accessing a package, importing packages.

UNIT-III :Exception handling-Benefits of exception handling, the classification of exceptions - exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception subclasses. Multithreading – Differences between multiple processes and multiple threads, thread life cycle, creating threads, interrupting threads, thread priorities, synchronizing threads, inter-thread communication, producer consumer problem.

UNIT-IV :Collection Framework in Java – Introduction to java collections, Overview of java collection framework, Commonly used collection classes- Array List, Vector, Hash table, Stack, Lambda Expressions. Files- Streams- Byte streams, Character streams, Text input/output, Binary input/output, File management using File class. Connecting to Database – JDBC Type 1 to 4 drivers, Connecting to a database, querying a database and processing the results, updating data with JDBC.

UNIT-V :GUI Programming with Swing - The AWT class hierarchy, Introduction to Swing, Swing Vs AWT, Hierarchy for Swing components, Overview of some Swing components – JButton, JLabel, JTextField, JTextArea, simple Swing applications, Layout management – Layout manager types – border, grid and flow Event Handling- Events, Event sources, Event classes, Event Listeners, Delegation event model, Examples: Handling Mouse and Key events, Adapter classes.

TEXT BOOK:

1. Java Fundamentals-Comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH.
2. Core Java: An Integrated Approach – Dr R Nageswara Rao

REFERENCE BOOKS:

1. Java for Programmers, P.J.Deitel and H.M.Deitel, PEA (or) Java: How to Program, P.J.Deitel and H.M.Deitel, PHI
2. Object Oriented Programming through Java, P.Radha Krishna, Universities Press.
3. Thinking in Java, Bruce Eckel, PE
4. Programming in Java, S. Malhotra and S. Choudhary, Oxford Universities Press.
5. Design Patterns Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides.

Course Outcome :

CO1	Explain Java syntax and OOP principles.	K2	IO
CO2	Develop Java programs using classes and objects.	K4,K5	HO
CO3	Apply inheritance, polymorphism, and abstraction.	K5	HO
CO4	Use exception handling and collections framework.	K1	LO
CO5	Build modular Java applications.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Object Oriented Programming through Java lab		
Elective - IV lab	I Year & II Semester	Credit: 3	

Course Objectives :

1. Enable students to implement core **object-oriented programming concepts** such as abstraction, encapsulation, inheritance, and polymorphism.
2. Develop skills in writing, compiling, debugging, and executing **Java programs**.
3. Provide hands-on experience with **classes, objects, and methods** in real-world problem solving.
4. Demonstrate the use of **packages, interfaces, and exception handling**.
5. Gain practical knowledge in **multithreading and synchronization**.
6. Work with **Java Collection Framework** and file handling techniques.
7. Build database-connected applications using **JDBC**.
8. Design **Graphical User Interfaces (GUI)** using Swing and handle events.

List of Experiments

1. Write simple Java programs using:

- Data types, variables, constants
- Operators and **control** structures (if, switch, loops)

2. Programs on:

- One-dimensional and two-dimensional arrays
- String manipulation methods

3. Create a class with:

- Fields, constructors, and methods
- Demonstrate object creation and method invocation

4. Implement:

- Constructor overloading
- Method overloading

5. Programs demonstrating:

- Single inheritance
- Multilevel inheritance
- Use of super keyword

6. Implement:

- Method overriding
- Runtime polymorphism

7. Create:

- Abstract classes and methods

8. Create and use:

- User-defined packages
- Importing and accessing packages

9. Programs using:

- try, catch, finally, throw, throws
- User-defined exceptions

10. Programs to demonstrate:

- Thread creation (Thread class & Runnable interface)
- Synchronization and thread communication

11. Implement:

- ArrayList, Vector, Stack, Hashtable
- Basic operations and iteration

12. Programs using:

- Byte streams and character streams
- Reading and writing files
- File class operations

13. Develop programs to:

- Connect to database
- Execute SELECT, INSERT, UPDATE queries

14. Create applications using:

- JLabel, JButton, JTextField, JTextArea
- Layout managers (Flow, Border, Grid)

15. Programs to handle:

- Mouse events
- Keyboard events
- Action events using listeners

Course Outcome :

CO1	Develop Java programs using core language constructs.	K2	IO
CO2	Implement object-oriented concepts such as inheritance and polymorphism.	K4,K5	HO
CO3	Apply exception handling and file processing in Java programs.	K5	HO
CO4	Use collections and GUI/database integration in Java applications.	K1	LO
CO5	Build modular and reusable Java-based software solutions.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Statistics for Data Science		
Core-VIII Theory	I Year & II Semester	Credit: 3	

Course Objective:

- To introduce fundamental concepts of statistics and data visualization techniques.
- To develop understanding of probability theory and statistical distributions.
- To enable students to perform statistical inference and hypothesis testing.
- To apply correlation, regression, and modeling techniques to real-world data.
- To expose students to advanced statistical methods and their applications in data science.
- To provide hands-on exposure to statistical tools such as Python and R

Course Outcomes

After completing this course, students will be able to:

1. Understand and apply fundamental statistical concepts in data analysis.
2. Analyze datasets using descriptive and inferential statistical methods.
3. Apply probability theory to model uncertainty in data science problems.
4. Perform hypothesis testing and interpret statistical results.
5. Use statistical tools and software for data-driven decision-making.
6. Build and evaluate statistical models for real-world applications.

UNIT I: Descriptive Statistics and Data Visualization

Introduction to Statistics in Data Science - Types of Data: Nominal, Ordinal, Interval, Ratio - Data Collection Methods and Sampling Techniques - Measures of Central Tendency: Mean, Median, Mode - Measures of Dispersion: Range, Variance, Standard Deviation - Skewness and Kurtosis - Data Visualization Techniques: Histograms, Bar Charts, Pie Charts - Box Plots, Scatter Plots - Introduction to statistical tools (Python/R basics)

UNIT II: Probability Theory and Distributions

Basics of Probability - Axioms of Probability - Conditional Probability and Bayes' Theorem - Random Variables: Discrete and Continuous - Probability Distributions: Binomial Distribution - Poisson Distribution - Normal Distribution - Central Limit Theorem - Law of Large Numbers

UNIT III: Inferential Statistics

Sampling Distributions - Estimation Techniques: Point Estimation - Interval Estimation - Confidence Intervals - Hypothesis Testing: Null and Alternative Hypothesis - Type I and Type II Errors - Parametric Tests: Z-test - t-test - Chi-square test - Non-parametric Tests (Basics)

UNIT IV: Correlation, Regression and Statistical Modeling

Correlation Analysis: Pearson and Spearman Correlation - Regression Analysis: Simple Linear Regression - Multiple Linear Regression - Model Assumptions and Evaluation - Residual Analysis - Introduction to Logistic Regression - Overfitting and Underfitting

UNIT V: Advanced Statistical Methods and Applications

Analysis of Variance (ANOVA) - Time Series Analysis: Trend, Seasonality - Introduction to Bayesian Statistics - Resampling Methods: Bootstrapping - Introduction to Statistical Learning - Applications in Data Science: Business Analytics - Healthcare Analytics - Social Media Data Analysis - Tools: Python (NumPy, Pandas, SciPy), R

Recommended Reference/Text Books:

1. Introduction to Probability and Statistics for Engineers and Scientists – Sheldon M. Ross, 5th Edition, Academic Press.
2. Statistics for Data Science – James D. Miller, Latest Edition, Packt Publishing.
3. An Introduction to Statistical Learning – Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, 2nd Edition, Springer.
4. Practical Statistics for Data Scientists – Peter Bruce, Andrew Bruce, 2nd Edition, O'Reilly Media.
5. The Elements of Statistical Learning – Trevor Hastie, Robert Tibshirani, Jerome Friedman, 2nd Edition, Springer.

Learning Resources:

Online Platforms

- Coursera – Statistics and Data Science Specializations
- edX – Probability and Statistics Courses
- Khan Academy – Fundamental statistics

Software Tools

- Python (NumPy, Pandas, Matplotlib, SciPy)
- R
- IBM SPSS

Datasets for Practice

- Kaggle
- UCI Machine Learning Repository

Course Outcome :

CO1	Understand, describe, analyze and examine descriptive statistics, data types, and data visualization techniques.	K1-K6
CO2	Comprehend and apply probability concepts, random variables, and probability distributions in data analysis	K1-K6
CO3	Understand, apply and evaluate statistical inference methods including estimation and hypothesis testing. .	K1-K6
CO4	Understand, analyze and evaluate correlation and regression models for data interpretation and prediction.	K1-K6
CO5	Comprehend, apply and appraise advanced statistical techniques and their applications in real-world data science problems.	K1-K6

Course Mapping :

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	L	L	M	-	-	-	L	M	-	M
CO2	S	S	L	M	M	-	-	-	L	M	-	M
CO3	S	S	M	S	M	-	-	-	L	M	-	M
CO4	S	S	S	S	S	-	-	-	M	M	L	M
CO5	S	S	S	S	S	L	L	L	M	M	L	M

SEMESTER III

Title of the Paper	Web Technology (React JS, Node JS, Mongo DB)		
Core-IX Theory	II Year & III Semester	Credit: 4	

Course Objective

- To know the concepts of JavaScript and Full Stack Technology.
- To explore the concepts of React JS.
- To know the concepts of Node JS.
- To know the concepts of mongo DB.
- To implement the deployment process.

UNIT I: Introduction -JavaScript -Execution Context and Call Stack - Hoisting in JavaScript - Spread Operator - Scope Chain, Temporal Dead Zone - Block Scope, Shadowing-Closures. Full Stack Technology

UNIT II: React.js :React Introduction - What is React - What is a Component - JSX Overview. create-react-app :Understanding basics of react app. Understanding virtual DOM, SPA. Components - Class Components - Functional Components -Parent, Child Components - Conditional Rendering - State, setState Method - Props. Event Handling in React: Event Handling in Class Components-Event Handling in Functional Components. Lifecycle : Class Component Life Cycle Methods. React Hooks: What is a React Hook - useState Hook - useEffect Hook .Building forms in React. React Router. Controlled vs Uncontrolled Components State Management: Single Source of Truth - Lifting State Up - Prop Drilling - useContext -Redux. HTTP Methods - Fetch - Axios .

UNIT III: Node.js & Express.js :Introduction to Node.js-Creating a Simple Server- Response types - HTML, JSON -Modules -NPM -Introduction to Express.js -Express Params and Query String -Express Router .

UNIT IV: Mongo DB: SQL/NoSQL landscape - Document Vs. Other types of Storage - MongoDB feature set - Introduction to BSON and JSON - Simple Queries.Connecting with Node JS: Inserts and Retrievals -Updates and Deletes

UNIT V: MERN & Deployment: Connecting React and Node - Building an application in MERN - Github and CI/CD - Deploy using Netlify / Heroku

Text Book

1. Eloquent JavaScript: A Modern Introduction to Programming, No Starch Press
2. You Don't Know JS, Shroff/O'Reilly
3. The Road to Learn React: Your Journey to Master Plain Yet Pragmatic React.js, Zaccheus Entertainment
4. React Explained: Your Step-by-Step Guide to React, OS Training, LLC
5. Beginning React, Greg Lim
6. Learning React: Functional Web Development with React and Redux, Shroff/O'Reilly
7. Learn React Hooks: Build and refactor modern React.js applications using Hooks, Packt Publishing Limited
8. Pro React, Apress
9. Web Development with Node and Express: Leveraging the JavaScript Stack, O'Reilly Media
10. Express in Action: Writing, building, and testing Node.js applications, Manning Publications
11. Beginning Node.js, Express & MongoDB Development, Greg Lim
12. MongoDB: The Definitive Guide - Powerful and Scalable Data Storage, Shroff/O'Reilly; Third edition
13. Full-Stack React Projects: Learn MERN stack development by building modern web apps using MongoDB, Express, React, and Node.js, Packt Publishing Limited
14. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Apress

Course Outcome :

CO1	Explain front-end and back-end web technologies.	K2	IO
CO2	Develop responsive web interfaces using modern frameworks.	K4,K5	HO
CO3	Implement server-side web applications and APIs.	K5	HO
CO4	Integrate databases with web applications.	K1	LO
CO5	Deploy and maintain full-stack web applications.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	S	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Advanced Machine Learning		
Core- X Theory	II Year & III Semester	Credit:5	535C3C

Course Objectives

- To understand the concepts of Machine Learning.
- To understand the theoretical and practical aspects of types of machine learning
- To teach and get familiarized with supervised learning and their applications.
- To teach and get familiarized with the concepts and algorithms of unsupervised learning.
- To appreciate the concepts and algorithms of deep learning.

UNIT I: Introducing Machine Learning: The Origins of Machine Learning, Uses and Abuses of Machine Learning _ Basics of Machine Learning Algorithm Model Works - Steps to apply Machine Learning - Choosing a Machine Learning Algorithm - Using Machine Learning concepts.

Managing and Understanding Data: Data Structures, Vectors And Factors: Lists, Data frames, Matrixes and arrays - Managing Data - Exploring and Understanding Data: Exploring the Structure of Data, Exploring Numeric variables - Exploring Categorical Variables- Exploring Relationships between Variables.

UNIT II: Lazy Learning – Classification Using Nearest Neighbors: The kNN Algorithm- Diagnosing Breast Cancer with the KNN Algorithm- Probabilistic Learning – Classification Using Naive Bayes: Basic concepts of Bayesian Methods- The Naïve Bayes Algorithm- Example – filtering Mobile Phone Spam with the Naive Bayes Algorithm.

Divide and Conquer – Classification Using Decision Trees and Rules: Understanding Decision Trees- Example – Identifying Risky Bank Loans using C5.0 Decision Trees- Understanding Classification Rules- Example – Identifying Poisonous Mushrooms with Rule Learners.

UNIT III: Forecasting Numeric Data – Regression Methods: Understanding Regression- Example – Predicting Medical Expenses using Linear Regression- Understanding Regression Trees and Model Trees- Example – Estimating the Quality of Wines with Regression Trees and Model Trees.

Black Box Methods Neural Networks and Support Vector Machines: Understanding Neural

Networks, from Biological to Artificial Neurons, Activation Functions, Network Topology, Training Neural Networks with Backpropagation - Modeling the Strength of Concrete with ANNs- Understanding Support Vector Machines- Performing OCR with SVMs- Finding Patterns -Market Basket Analysis Using Association Rules: Understanding Association Rules- Example Identifying Frequently Purchased Groceries with Association Rules.

UNIT IV: Finding Groups of Data – Clustering with K-Means: Understanding Clustering- The k-means Algorithm for clustering- Finding teen market segments using k-means Clustering- Evaluating Model Performance: Measuring Performance for Classification- Beyond Accuracy – other Measures of Performance, Visualizing Performance Tradeoffs.

Improving Model Performance: Tuning Stock Models for Better Performance-Using Caret for Automated Parameter Tuning- Creating a simple Tuned Model- Customizing the Tuning Process- Improving Model Performance with meta-learning- Understanding Ensembles- Bagging- Boosting- Random forests.

UNIT V: Introduction to Deep Learning: Introduction to Deep Learning, Single Layer Perceptron Model (SLP), Multilayer Perceptron Model (MLP), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Restricted Boltzmann Machines (RBMs).

Convolutional Neural Networks (CNNs): Structure and Properties of CNNs - Components of CNN Architectures- Convolutional Layer, Pooling Layer, Rectified Linear Units (ReLU) Layer, Fully Connected (FC) Layer, Loss Layer - Tuning Parameters ,Notable CNN Architectures, Regularization- Recurrent Neural Networks (RNNs): Fully Recurrent Networks, Training RNNs with Back-Propagation Through Time (BPPT)- Elman Neural Networks, Neural History Compressor, Long Short-Term Memory (LSTM), Traditional and Training LSTMs - Structural Damping Within RNNs, Tuning Parameter Update Algorithm.

Text Books:

1. Brett Lantz, “Machine Learning with R”, Addison-Wesley Packt Publishing, 2013.
2. TawehBeysolow, “Introduction to Deep Learning Using R: A Step-by-Step Guide to Learning and Implementing Deep Learning Models Using R”, San Francisco, California, USA, 2017.

Reference Books:

1. Daniel T. Larose, Chantal D. Larose, “Data mining and Predictive analytics”, Second Ed., Wiley Publication, 2015.
2. Bertt Lantz, “Machine Learning with R: Expert techniques for predictive modeling”, 3rd Edition, April 15,2019,
3. Jason Bell, “Machine Learning: Hands-On for Developers and Technical Professionals”, Wiley Publication,2015.

Course Outcome :

CO1	Explain machine learning workflow and algorithms.	K2	IO
CO2	Apply supervised learning techniques to real datasets.	K4,K5	HO
CO3	Build classification and regression models.	K5	HO
CO4	Evaluate and optimize machine learning models.	K1	LO
CO5	Develop deep learning solutions for advanced tasks.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	Data Science and Analytics		
Core XI - Theory	II Year & III Semester	Credit: 3	

COURSE OBJECTIVES:

- Understand the Big Data Platform and its Use cases
- Provide an overview of Apache Hadoop
- Provide HDFS Concepts and Interfacing with HDFS
- Understand Map Reduce Jobs
- Provide hands on Hadoop Eco System

UNIT I: Introduction to big data and hadoop :Types of Digital Data, Introduction to Big Data, Big Data Analytics, History of Hadoop, Apache Hadoop, AnalysingData with Unix tools, Analysing Data with Hadoop, Hadoop Streaming, Hadoop Echo System, IBM Big DataStrategy, Introduction to Infosphere BigInsights and Big Sheets.

UNIT II:HDFS(Hadoop Distributed File System) The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, DataIngest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization, Avro and File-BasedData structures

UNIT III:Map Reduce: Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Typesand Formats, Map Reduce Features.

UNIT IV: Hadoop Eco System: Pig : Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User DefinedFunctions, Data Processing operators. Hive : Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, QueryingData and User Defined Functions.

UNIT V: Hbase: HBasics, Concepts, Clients, Example, Hbase Versus RDBMS.Big SQL: Introduction - Machine Learning: Introduction, Supervised Learning, Unsupervised Learning, Collaborative Filtering.

Text Books

1. Tom White “ Hadoop: The Definitive Guide” Third Edit on, O’reily Media, 2012.
2. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015

References

1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.
2. Jay Liebowitz, "Big Data and Business Analytics" Auerbach Publications, CRC press (2013)
3. Anand Rajaraman and Jeffrey David Ullman, "Mining of Massive Datasets", Cambridge University Press, 2012.

Course Outcome :

CO1	Explain big data concepts and Hadoop ecosystem.	K2	IO
CO2	Use HDFS and Hadoop I/O mechanisms.	K4,K5	HO
CO3	Develop MapReduce programs for analytics.	K5	HO
CO4	Apply Hive and Pig for big data processing.	K1	LO
CO5	Integrate big data tools with analytics pipelines.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	M	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Course/ Paper	Web Technology Lab		
Core-xii	II Year & III Semester	Credit : 3	

Coursre Objectives:

- To implement the client side of the web application using javascript. □
- To understand Javascript on the desktop using NodeJS.
- To develop a web application using NodeJS and Express.
- To implement a SPA using React. □
- To develop a full stack single page application using React, NodeJS, and a Database (MongoDB or SQL).

List of Experiments:

1. Use table tag to format web page. Also create the Time-Table of your class using table tag.
2. Create your profile page i.e. educational details, Hobbies, achievement, My Ideals etc.
3. Create Style sheet to set formatting for text tags and embed that style sheet on web pages created for your site.
4. Design a web page and embed various multimedia features in the page.
5. Design signup form to validate username, password, and phone numbers etc using Java script.
6. Write a JavaScript program to determine whether a given year is a leap year in the Gregorian calendar.
7. Write a JavaScript program to convert temperatures to and from celsius, Fahrenheit.
8. Create a counter using ReactJS.
9. Create a Todo application using ReactJS.

Course Outcome :

CO1	Build responsive front-end web pages.	K2	IO
CO2	Develop dynamic client-side scripts.	K4,K5	HO
CO3	Create backend-integrated web applications.	K5	HO
CO4	Connect web apps with databases/APIs.	K1	LO
CO5	Test and deploy web applications.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	S	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Advanced Machine Learning Lab (AML)		
NME - II Practical - VI	II Year & III Semester	Credit:2	535S3A

Course Objectives

- To formulate machine learning problems corresponding to different applications.
- To understand a range of machine learning algorithms along with their strengths and weaknesses.
- To apply machine learning algorithms to solve problems of moderate complexity. To apply CNN to solve problems of moderate complexity.
- To apply LSTM and RNN to solve problems.

List of Programs

1. Write a python program to compute the Central Tendency Measures: Mean, Median, Mode, Measure of Dispersion: Variance, Standard Deviation
2. Implement a Linear Regression and Multiple Linear Regression with a Real Dataset
3. Implementation of Logistic Regression using sklearn
4. Implement a binary classification model.
5. Classification with Nearest Neighbours and NavieBayes Algorithm
6. Implementation Decision tree for classification using sklearn and its parameter tuning
7. Implement the k-means algorithm.
8. Implement an Image Classifier using CNN in TensorFlow/Keras.
9. Implement an Autoencoder in TensorFlow/Keras.
10. Implement a Simple LSTM using TensorFlow/Keras.

Course Outcome :

CO1	Implement supervised learning algorithms practically.	K2	IO
CO2	Build regression and classification models.	K4,K5	HO
CO3	Evaluate models using performance metrics.	K5	HO
CO4	Tune hyperparameters and optimize models.	K1	LO
CO5	Develop deep learning prototypes for real-world tasks.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	M	S	-	-	-	-	-	-
CO2	S	S	-	S	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	S	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	Cloud Computing		
Elective -V Theory	II Year & III Semester	Credit:3	535E3C

Course Objectives:

- To introduce the cloud computing concepts and map reduce programming model.
- To provide skills and knowledge about operations and management in cloud technologies so as to implement large scale systems.
- To provide skills to design suitable cloud infrastructure that meets the business services and customer needs.
- To provide Knowledge of different CPU, memory and I/O virtualization techniques that serve in offering software, computation and storage services on the cloud;

UNIT I: Foundations of cloud: Inception and need for cloud computing: Motivations from distributed computing predecessors - Evolution - Characteristics - Business Benefits – Challenges in cloud computing - Exploring the Cloud Computing Stack - Fundamental Cloud Architectures – Advanced Cloud Architectures - Specialized Cloud Architectures.

UNIT II:Service Delivery and Deployment Models: Service Models (XaaS): Infrastructure as a Service (IaaS) - Platform as a Service (PaaS) - Software as a Service(SaaS) - Deployment Models: Types of cloud - Public cloud - Private cloud - Hybrid cloud – Service level agreements- Types of SLA – Lifecycle of SLA- SLA Management

UNIT III: Cloud Resource Virtualization: Virtualization as Foundation of Cloud – Understanding Hypervisors – Understanding Machine Image and Instances - Managing Instances – Virtual Machine Provisioning and Service Migrations Cloud Computing Applications and Paradigms: Existing Cloud Applications and Opportunities for New Applications - Architectural Styles for Cloud Applications - Workflows: Coordination of Multiple Activities - Coordination Based on a State Machine Model: The ZooKeeper - The MapReduce Programming Model - A Case Study: The Grep The Web Application

UNIT IV : Resource Management and Scheduling in Cloud: Policies and Mechanisms for Resource Management – Stability of a Two-Level Resource Allocation Architecture-Feedback

Control Based on Dynamic Thresholds - Coordination of Specialized Autonomic Performance Managers - A Utility-Based Model for Cloud-Based Web Services - Resource Bundling: Combinatorial Auctions for Cloud Resources – Scheduling Algorithms for Computing Clouds - Resource Management and Dynamic Application Scaling

UNIT V : Cloud Platforms and Application Development: Comparing Amazon web services, Google AppEngine, Microsoft Azure from the perspective of architecture (Compute, Storage Communication) services and cost models. Cloud application development using third party APIs, Working with EC2 API – Google App Engine API - Facebook API, Twitter API. Advances in Cloud: Media Clouds - Security Clouds - Computing Clouds - Mobile Clouds – Federated Clouds – Hybrid Clouds

Text Books:

1. Rajkumar Buyya, James Broberg, Andrzej, M. Goscinski, Cloud Computing: Principles and Paradigms, Wiley, 1st Edition, 2013.
2. Sosinski, Barrie, Cloud Computing Bible, John Wiley & Sons, 1st Edition, 2011.

Reference Books:

1. Marinescu, Dan C. Cloud Computing: Theory and Practice. Morgan Kaufmann, 2017.
2. Toby Velte, Anthony Velte, Robert Elsenpeter, Cloud Computing: A Practical Approach, Mc Graw Hill Education, 1st Edition, 2017.
3. Buyya, Rajkumar, Christian Vecchiola, and S. Thamarai Selvi. Mastering Cloud Computing: Foundations and Applications Programming, Tata Mcgraw Hill, 1st Edition, 2017.

Web References:

1. <https://www.youtube.com/watch?v=-8O32k26RWA>

Course Outcome :

CO1	Explain cloud computing models and architecture.	K2	IO
CO2	Use virtualization and cloud deployment tools.	K4,K5	HO
CO3	Develop cloud-based applications/services.	K5	HO
CO4	Implement cloud security and monitoring practices.	K1	LO
CO5	Evaluate cloud solutions for enterprise needs.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	M	S	M	S	-	-	-	S	-	-
CO5	S	S	M	S	S	S	-	-	-	-	-	-

Title of the Paper	Software Development Technologies		
Elective - V Theory	II Year & III Semester	Credit:3	535E3A

Course Objectives

- To learn and Implementing Micro services
- To analysing the Azure Kubernetes Service
- To learn and anlyse .NET DevOps for Azure and its applications
- To building code for .NET core applications
- To get familiarized with Azure pipelines

UNIT I: Implementing Micro services: Client to microservices communication, Interservice communication, data considerations, security, monitoring, microservices hosting platform options. Azure Service Fabric: Introduction, core concepts, supported programming models, service fabric clusters, develop and deploy applications of service fabric. Monitoring Azure Service Fabric Clusters: Azure application, resource manager template, Adding Application Monitoring to a Stateless Service Using Application Insights, Cluster monitoring, Infrastructure monitoring.

UNIT II: Azure Kubernetes Service (AKS): Introduction to kubernetes and AKS, AKS development tools, Deploy applications on AKS. Monitoring AKS: Monitoring, Azure monitor and analytics, monitoring AKS clusters, native kubernetes dashboard, Prometheus and Grafana. Securing Microservices: Authentication in microservices, Implenting security using API gateway pattern, Creating application using Ocrlot and securing APIs with Azure AD. Database Design for Microservices: Data stores, monolithic approach, Microservices approach, harnessing cloud computing, database options on MS Azure, overcoming application development challenges. Building Microservices on Azure Stack: Azure stack, Offering IaaS, PaaS on-premises simplified, SaaS on Azure stack.

UNIT III : .NET DevOps for Azure: DevOps introduction, Problem and solution. Professional Grade DevOps Environment: The state of DevOps, professional grade DevOps vision, DevOps

architecture, tools for professional DevOps environment, DevOps centered application. Tracking work: Process template, Types of work items, Customizing the process, Working with the process. Tracking code: Number of repositories, Git repository, structure, branching pattern, Azure repos configuration, Git and Azure.

UNIT IV: Building the code: Structure of build, using builds with .NET core and Azure pipelines, Validating the code: Strategy for defect detection, Implementing defect detection. Release candidate creation: Designing release candidate architecture, Azure artifacts workflow for release candidates, Deploying the release: Designing deployment pipeline, Implementing deployment in Azure pipelines. Operating and monitoring release: Principles, Architectures for observability, Jumpstarting observability.

UNIT V: Introduction to APIs: Introduction, API economy, APIs in public sector. API Strategy and Architecture: API Strategy, API value chain, API architecture, API management. API Development: Considerations, Standards, kick-start API development, team orientation. API Gateways: API Gateways in public cloud, Azure API management, AWS API gateway. API Security: Request-based security, Authentication and authorization.

Text Books

1. Harsh Chawla and Hemant Kathuria, Building Microservices Applications on Microsoft Azure- Designing, Developing, Deploying, and Monitoring, Apress, 2019.
2. Jeffrey Palermo , NET DevOps for Azure A Developer's Guide to DevOps Architecture the Right Way, Apress, 2019.
3. Thurupathan and Vijayakumar, Practical API Architecture and Development with Azure and AWS - Design and Implementation of APIs for the Cloud, Apress, 2018.

Reference Books

1. Karl Matthias and Sean P. Kane, Docker: Up and Running, O'Reilly Publication, Second Edition 2018.
2. Len Bass, Ingo Weber, Liming Zhu, "DevOps, A Software Architects Perspective", Addison Wesley-Pearson Publication, First Edition 2015.
3. John Ferguson Smart, "Jenkins, The Definitive Guide", O'Reilly Publication, First Edition 2011.

Course Outcome :

CO1	Explain modern software development methodologies.	K2	IO
CO2	Apply version control and collaboration workflows.	K4,K5	HO
CO3	Use development frameworks and build tools.	K5	HO
CO4	Implement CI/CD and DevOps practices.	K1	LO
CO5	Manage software projects using agile practices.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	M	M	-	S	-	-	-	-	-	-
CO2	S	S	M	M	M	S	-	-	-	-	-	-
CO3	S	S	M	S	M	S	-	-	-	-	-	-
CO4	S	S	M	S	S	S	-	-	-	-	-	-
CO5	S	S	M	S	S	S	-	-	-	-	-	-

Title of the Paper	Social Network Analysis		
Elective - V Theory	II Year & III Semester	Credit:3	535E3A

Course objectives:

- To introduce the fundamentals of Social Network Analysis and its significance in understanding societal connections and behaviors.
- To analyze various models of network growth and understand the properties of real-world networks.
- To explore link analysis algorithms and their applications in understanding relationships within a network.
- To study community detection methods and their relevance in identifying meaningful clusters within networks.
- To understand link prediction techniques and their application in forecasting future connections within a network.

UNIT I : Networks and Society - What is Social Network Analysis, why do We Study Social Networks, Applications of Social Network Analysis, Preliminaries, Three Levels of Social Network Analysis.

Network Measures - Network Basics, Node Centrality, Assortativity, Transitivity and Reciprocity, Similarity, Degeneracy.

UNIT II : Network Growth Models - Properties of Real-World Networks, Random Network Model, Ring Lattice Network Model, Watts–Strogatz Model, Preferential Attachment Model, Price’s Model, Local-world Network Growth Model, Network Model with Accelerating Growth, Aging in Preferential Attachment.

UNIT III : Link Analysis - Applications of Link Analysis, Signed Networks, Strong and Weak Ties, Link Analysis Algorithms, PageRank, Personalised PageRank, DivRank, SimRank, PathSIM.

UNIT IV : Community Structure in Networks - Applications of Community Detection, Types of Communities, Community Detection Methods, Disjoint Community Detection, Overlapping Community Detection, Local Community Detection, Community Detection vs Community Search, Evaluation of Community Detection Methods.

UNIT V: Link Prediction - Applications of Link Prediction, Temporal Changes in a Network, Problem Definition Evaluating Link Prediction Methods, Heuristic Models, Probabilistic Models, Supervised Random Walk, Information-theoretic Model, Latest Trends in Link Prediction.

Text Books

1. Tanmoy Chakraborty, “Social Network Analysis”, Wiley India Pvt. Ltd., 2021

Reference Books

1. Albert-Laszlo Barabasi, “Network Science”, Cambridge University Press, 2016
2. Stanley Wasserman, Katherine Faust, “Social Network Analysis: Methods and Applications”, Cambridge University Press, 1994

Course Outcome :

CO1	Explain social network concepts and graph models.	K2	IO
CO2	Analyze network structures and metrics.	K4,K5	HO
CO3	Apply community detection and centrality algorithms.	K5	HO
CO4	Model influence propagation in networks.	K1	LO
CO5	Use analytics tools for social network mining.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

SEMESTER IV

Title of the Paper	Data Ethics, Privacy and Governance		
Elective - V Theory	II Year & IV Semester	Credit:3	

Course Objectives:

- To introduce ethical issues in data collection and analysis
- To understand privacy-preserving techniques
- To familiarize students with legal frameworks
- To develop skills in data governance and compliance
- To analyze ethical challenges in AI systems

UNIT I: Introduction to Data Ethics-Definition and importance of data ethics - Ethical principles: fairness, accountability, transparency - Data lifecycle and ethical considerations - Bias in datasets and algorithms - Ethical decision-making models - Case studies

UNIT II: Data Privacy Fundamentals - Privacy concepts and personal data - Types of data and privacy risks - Data breaches and re-identification - Privacy-preserving techniques: anonymization, pseudonymization - Differential privacy basics - Privacy by design

UNIT III: Legal and Regulatory Frameworks - Overview of GDPR and Indian DPDP Act - Data subject rights - Roles: data controller and processor - Compliance requirements - Data protection impact assessment - Cross-border data flow

UNIT IV: Data Governance - Data governance concepts and frameworks - Data quality management - Data stewardship and ownership - Risk management and compliance - Governance tools and standards

UNIT V: Ethical Issues in AI and Data Science -Bias and fairness in AI - Explainable AI (XAI) - Accountability in automated systems - Responsible AI frameworks - Case studies: surveillance, recommendation systems

Textbooks:

1. Hasselbalch, G., & Tranberg, P., *Data Ethics of Power*, 1st Ed., Edward Elgar Publishing, 2023.
2. O'Neil, C., *Weapons of Math Destruction*, 1st Ed., Crown Books, 2016.
3. Oliver, I., *Privacy Engineering*, 1st Ed., McGraw-Hill Education, 2014.

Reference Book:

1. Ladley, J., *Data Governance*, 1st Ed., Morgan Kaufmann, 2012.
2. Floridi, L., *The Ethics of Information*, Reprint Ed., Oxford University Press, 2015.
3. Russell, S., & Norvig, P., *Artificial Intelligence: A Modern Approach*, 4th Ed., Pearson, 2020.

E-Learning Resources

- NPTEL Courses on Data Ethics & Privacy
- Coursera / edX Data Governance courses
- OECD AI Principles
- Government of India DPDP Act portal

Course Outcome :

CO1	Explain ethical principles in data handling.	K2	IO
CO2	Apply privacy-preserving techniques in data systems.	K4,K5	HO
CO3	Interpret legal and regulatory frameworks.	K5	HO
CO4	Develop governance strategies for data management.	K1	LO
CO5	Evaluate ethical challenges in AI and analytics.	K5	HO

Course Mapping :

CO/P O	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2
CO1	M	M	S	M	-	M	-	-	-	S	-	-
CO2	M	S	S	M	-	M	-	-	-	S	-	-
CO3	M	S	S	M	-	M	-	-	-	S	-	-
CO4	S	S	S	M	-	M	-	-	-	S	-	-
CO5	S	S	S	S	-	M	-	-	-	S	-	-

Title of the Paper	Project viva voce		
	II YEAR IV Semester	Credit: 4	

Title of the Paper	Natural Language Processing		
ELECTIVE	I Year & III Semester	Credit: 4	

Course Objectives:

- To learn the fundamentals of natural language processing.
- To understand the use of CFG and PCFG in NLP.
- To understand the role of semantics of sentences and pragmatics.
- To apply the NLP techniques to IR applications.

Course Outcomes:

- To tag a given text with basic Language features.
- To design an innovative application using NLP components.
- To implement a rule-based system to tackle morphology/syntax of a language.
- To design a tag set to be used for statistical processing for real-time applications.
- To compare and contrast the use of different statistical approaches for different types of NLP applications.

UNIT I: Introduction to NLP: What is NLP? Why NLP is Difficult? History of NLP, Advantages of NLP, Disadvantages of NLP, Components of NLP, Applications of NLP, How to build an NLP pipeline? Phases of NLP, NLP APIs, NLP Libraries

UNIT II: Language Modeling and Part of Speech Tagging: Unigram Language Model, Bigram, Trigram, N-gram, Advanced smoothing for language modeling, Empirical Comparison of Smoothing Techniques, Applications of Language Modeling, Natural Language Generation, Parts of Speech Tagging, Morphology, Named Entity Recognition

UNIT III: Words and Word Forms: Bag of words, skip-gram, Continuous Bag-Of-Words, Embedding representations for words Lexical Semantics, Word Sense Disambiguation, Knowledge Based and Supervised Word Sense Disambiguation.

UNIT IV: Text Analysis, Summarization and Extraction: Sentiment Mining, Text Classification, Text Summarization, Information Extraction, Named Entity Recognition, Relation Extraction, Question Answering in Multilingual Setting; NLP in Information Retrieval, Cross-Lingual IR.

UNIT V: Machine Translation: Need of MT, Problems of Machine Translation, MT Approaches, Direct Machine Translations, Rule-Based Machine Translation, Knowledge Based MT System, Statistical Machine Translation (SMT), Parameter learning in SMT (IBM models) using EM), Encoder-decoder architecture, Neural Machine Translation.

Text Books :

1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition Jurafsky, David, and James H. Martin, PEARSON.
2. Foundations of Statistical Natural Language Processing, Manning, Christopher D., and Hinrich Schütze, Cambridge, MA: MIT Press.

Reference Books:

1. Natural Language Understanding, James Allen. The Benjamin/Cummings Publishing Company Inc..
2. Natural Language Processing with Python – Analyzing Text with the Natural Language Tool kit Steven Bird, Ewan Klein, and Edward Loper.

Course Outcome :

CO1	Explain NLP pipeline and preprocessing methods.	K2	IO
CO2	Build tagging and language modelling systems.	K4,K5	HO
CO3	Apply semantic representation and embeddings.	K5	HO
CO4	Implement NLP applications for text analytics.	K1	LO
CO5	Design translation and summarization systems.	K5	HO

Course Mapping :

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	M	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	M	S	-	-	-	-	-	-

Title of the Paper	Computer Vision		
Core -XIII Theory	II Year & IV Semester	Credit:4	

Course Objectives:

- To understand the fundamental concepts related to Image formation and processing.
- To learn feature detection, matching and detection
- To become familiar with feature based alignment and motion estimation
- To develop skills on 3D reconstruction
- To understand image based rendering and recognition

UNIT I: Basic Image Handling and Processing: PIL – the Python Imaging Library- Matplotlib - NumPy - SciPy- Advanced example: Image de-noising. Local Image Descriptors: Harris corner detector-SIFT - Scale-Invariant Feature Transform-Matching Geotagged Images.

UNIT II: Image to Image Mappings:Homographies-Warping images-Creating Panoramas. Camera Models and Augmented Reality: The Pin-hole Camera Model-Camera Calibration-Pose Estimation from Planes and Markers-Augmented Reality.

UNIT III: Multiple View Geometry:Epipolar Geometry-Computing with Cameras and 3D Structure-Multiple View Reconstruction-Stereo Images. Clustering Images: K-means Clustering-Hierarchical Clustering-Spectral Clustering.

UNIT IV: Searching Images: Content based Image Retrieval-Visual Words-Indexing Images-Searching the Database for Images-Ranking Results using Geometry-Building Demos and Web Applications. Classifying Image Content: K-Nearest Neighbors-Bayes Classifier-Support Vector Machines-Optical Character Recognition.

UNIT V: Image Segmentation: Graph Cuts-Segmentation using Clustering-Variational Methods. OpenCV: Python Interface-OpenCV Basics-Processing Video-Tracking.

Text Books:

1. Richard Szeliski, “Computer Vision: Algorithms and Applications”, Springer-Texts in Computer Science, Second Edition, 2022.
2. Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, Second Edition, 2015

Reference Books

1. Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Second Edition, Cambridge University Press, March 2004.
2. Christopher M. Bishop; Pattern Recognition and Machine Learning, Springer, 2006
3. E. R. Davies, Computer and Machine Vision, Fourth Edition, Academic Press, 2012

Course Outcome :

CO1	Explain image processing and computer vision fundamentals.	K2	IO
CO2	Apply feature extraction and image matching techniques.	K4,K5	HO
CO3	Implement image alignment and motion estimation methods.	K5	HO
CO4	Perform image classification and retrieval tasks.	K1	LO
CO5	Develop segmentation and object tracking solutions.	K5	HO

Course Mapping :

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2
CO1	S	M	-	M	-	S	-	-	-	-	-	-
CO2	S	S	-	M	M	S	-	-	-	-	-	-
CO3	S	S	-	S	M	S	-	-	-	-	-	-
CO4	S	S	-	S	M	S	-	-	-	-	-	-
CO5	S	S	-	S	S	S	-	-	-	-	-	-

Title of the Paper	Parallel and Distributed Computing		
Elective- VI Theory	II Year & IV Semester	Credit:3	

Course Objectives:

- To analyze the current popular distributed systems such as peer-to-peer (P2P) systems will also be analyzed.
- To learn and apply knowledge of parallel and distributed computing techniques and methodologies.
- To learn the architecture and parallel programming in graphics processing units (GPUs).
- To understand the memory hierarchy and cost-performance tradeoffs.
- To gain experience in the design, development, and performance analysis of parallel and distributed applications

UNIT I: Introduction to Parallel Computing: The Idea of Parallelism, Power and potential of parallelism, examining sequential and parallel programs, Scope and issues of parallel and distributed computing, Goals of parallelism, Parallelism and concurrency using multiple instruction streams.

UNIT II: Parallel Architecture: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture, Architectural classification schemes, Memory access classification, Memory Issues: Shared vs. distributed, Symmetric multiprocessing (SMP), SIMD, Vector processing, GPU co-processing, Flynn's Taxonomy, Instruction Level support for parallel programming, Multiprocessor caches and Cache Coherence, Non-Uniform Memory Access (NUMA).

UNIT III: Parallel Algorithm Design Principles and Programming: Need for communication and coordination/synchronization, Scheduling and contention, Independence and partitioning, Task-Based Decomposition, Data Parallel Decomposition, Characteristics of task and interaction, Load balancing, Data Management, parallel algorithm models, Sources of overhead in parallel programs, Performance metrics for parallel algorithm implementations, Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort, Parallel graph Algorithms.

UNIT IV: Architectures Of Distributed Systems - Architectural Styles – System Architectures - Architectures Versus Middleware - Self-Management In Distributed Systems - Processes - Threads - Virtualization - Clients -Servers - Communication - Remote Procedure Call - Message-Oriented Communication - Stream-Oriented Communication - Multicast Communication

UNIT V: Distributed Object Based Systems - Architecture - Processes - communication - Naming - Synchronization - Fault Tolerance - Security - Distributed System Examples - File Systems And Web Based Systems

Text Books:

1. Ananth Grama, Anshul Gupta, and George Karypis, Vipin Kumar; Introduction to Parallel Computing; Addison Wesley; 2nd Edition; 2003.
2. A.S. Tanenbaum; Distributed Operating Systems; Create Space Independent Publishing Platform; 3rd edition; 2017

Reference Books:

1. Introduction To Parallel Programming, Steven Brawer, Academic Press
2. Introduction To Parallel Processing, M. Sasikumar, Dinesh Shikhare and P. Ravi Prakash, PHI
3. Randy Chow, T. Johnson, Distributed Operating Systems and Algorithms, Addison Wesley
4. Ian Foster: Designing and Building Parallel Programs – Concepts and tools for Parallel Software Engineering, Pearson Publisher, 1st Edition, 2019.
5. Parallel Programming in C with MPI and OpenMP Michael J. Quinn, McGrawHill Higher Education.

Web References:

1. <https://www.youtube.com/watch?v=qbQCQ0U6H0o&list=PLbMVogVj5nJQRvzE NlvMKA9q70ScSRZBQ>

Course Outcome :

CO1	Explain parallel and distributed computing models.	K2	IO
CO2	Design parallel algorithms for computational problems.	K4,K5	HO
CO3	Implement distributed systems and communication protocols.	K5	HO
CO4	Evaluate synchronization and fault tolerance methods.	K1	LO
CO5	Optimize applications for scalable distributed execution.	K5	HO

Course Mapping :

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2
CO 1	S	M	-	M	-	S	-	-	-	-	-	-
CO 2	S	S	-	M	M	S	-	-	-	-	-	-
CO 3	S	S	-	S	M	S	-	-	-	-	-	-
CO 4	S	S	M	S	M	S	-	-	-	-	-	-
CO 5	S	S	M	S	S	S	-	-	-	-	-	-